Problem 1. *We observe a random sequence* $\{y_i, x_i\}_{i=1}^n$. *Suppose the regression function* $E[y|x]$ *is given by a nonlinear parametric function* $m(x, \theta_0)$, $\theta_0 \in \Theta$, *that is,*

$$E[y|x] = m(x, \theta_0).$$

Suppose (i) $\{(x_i, y_i)\}_{i=1}^n$ *are IID; (ii)* $\Pr\{m(x_i, \theta) \neq m(x_i, \theta_0)\} > 0$ *for every* $\theta \neq \theta_0$, $\theta \in \Theta$; *(iii)* $E \sup_{\theta \in \Theta}(y - m(x_i, \theta))^2 < \infty$; *(iv)* $m(x_i, \theta)$ *is continuous at each* $\theta \in \Theta$ *with probability 1; (v)* Θ *is compact. Show that the nonlinear least square estimator is consistent.*

Solution.

Let $Q_n(\theta) = \frac{1}{n}\sum_{i=1}^n (y_i - m(x_i, \theta))^2$ denote the sample objective function and $Q(\theta) = E[(y - m(x, \theta))^2]$ the population objective function. Since $E[y|x] = m(x, \theta_0)$, write $y = m(x, \theta_0) + \epsilon$ with $E[\epsilon|x] = 0$. Expanding the population objective, $Q(\theta) = E[(y - m(x, \theta))^2]$ $= E[(m(x, \theta_0) + \epsilon - m(x, \theta))^2] = E[(m(x, \theta_0) - m(x, \theta))^2] + 2E[\epsilon(m(x, \theta_0) - m(x, \theta))] + E[\epsilon^2]$. By the law of iterated expectations, $E[\epsilon(m(x, \theta_0) - m(x, \theta))] = E[E[\epsilon|x](m(x, \theta_0) - m(x, \theta))] = 0$, so $Q(\theta) = E[(m(x, \theta_0) - m(x, \theta))^2] + E[\epsilon^2]$. For any $\theta \neq \theta_0$, condition (ii) ensures that $\Pr\{m(x, \theta) \neq m(x, \theta_0)\} > 0$, which implies $E[(m(x, \theta_0) - m(x, \theta))^2] > 0$. Thus $Q(\theta) > Q(\theta_0)$ for all $\theta \neq \theta_0$, establishing that θ_0 uniquely minimizes $Q(\theta)$.

To show consistency, verify uniform convergence of the sample objective function. Define $g(z, \theta) = (y - m(x, \theta))^2$ with $z = (x, y)$. By condition (iv), $m(x, \theta)$ is continuous in θ with probability 1, so $g(z, \theta)$ is continuous in θ for almost every z. Condition (iii) provides $E[\sup_{\theta \in \Theta} |g(z, \theta)|] = E[\sup_{\theta \in \Theta}(y - m(x, \theta))^2] < \infty$, the uniform integrability condition. Combined with compactness of Θ (condition (v)) and the IID assumption (condition (i)), the uniform law of large numbers (Newey's ULLN for dominated functions over compact parameter spaces) yields $\sup_{\theta \in \Theta} |Q_n(\theta) - Q(\theta)| \xrightarrow{p} 0$. Since $Q_n(\theta)$ converges uniformly to $Q(\theta)$ over compact Θ and θ_0 uniquely minimizes $Q(\theta)$, the consistency theorem for extremum estimators implies $\hat{\theta} \xrightarrow{p} \theta_0$. □

Problem 2. *Consider the following probit model*

$$y_i = \mathbf{1}\{x_i'\beta_0 + \epsilon_i > 0\}, \quad \epsilon_i | x_i \sim N(0, 1), \quad \beta_0 \in \Theta \subset \mathbb{R}^k.$$

We observe $\{(x_i, y_i)\}_{i=1}^n$. *Even though ML is the way to go for this model, here we consider NLLS.*

(i) *Obtain the regression function* $m(x, \beta) = E[y_i|x_i]$.

(ii) *Using the function obtained in (i), consider NLLS that minimizes* $Q_n(\beta) = \sum_{i=1}^n (y_i - m(x, \beta))^2$ *over* $\theta \in \Theta$. *Suppose (a)* $\{(x_i, y_i)\}_{i=1}^n$ *are IID; (b)* Θ *is compact; (c)* $E[x_i x_i'] < \infty$ *is nonsingular. Use the result from the previous question to show that the NLLS estimator is consistent.*

Solution.

The regression function is $m(x, \beta) = E[y|x] = E[\mathbf{1}\{x'\beta_0 + \epsilon > 0\}|x] =$
$P(x'\beta_0 + \epsilon > 0|x) = P(\epsilon > -x'\beta_0|x) = P(\epsilon > -x'\beta_0) = 1 - \Phi(-x'\beta_0) = \Phi(x'\beta_0)$,
where Φ denotes the standard normal CDF. For a general β, $m(x, \beta) = \Phi(x'\beta)$.

To apply Problem 1, verify the five conditions with θ replaced by β. Condition (i) holds by assumption (a). For condition (ii), need $\Pr\{\Phi(x_i'\beta) \neq \Phi(x_i'\beta_0)\} > 0$ for $\beta \neq \beta_0$. Since Φ is strictly increasing, $\Phi(x_i'\beta) \neq \Phi(x_i'\beta_0)$ if and only if $x_i'\beta \neq x_i'\beta_0$, which occurs if and only if $x_i'(\beta - \beta_0) \neq 0$. For $\beta \neq \beta_0$, the vector $\beta - \beta_0 \neq 0$. Since $E[x_i x_i']$ is nonsingular (assumption (c)), the distribution of x_i is not concentrated on any proper subspace of $\mathbb{R}^k$, hence $\Pr\{x_i'(\beta - \beta_0) \neq 0\} > 0$. For condition (iii), since $y_i \in \{0, 1\}$ and $\Phi(x_i'\beta) \in [0, 1]$, have $(y_i - \Phi(x_i'\beta))^2 \leq 1$ for all $\beta \in \Theta$, so $E \sup_{\beta \in \Theta}(y_i - \Phi(x_i'\beta))^2 \leq 1 < \infty$. Condition (iv) holds since $\Phi(\cdot)$ is continuous and $x_i'\beta$ is continuous in β, making $m(x_i, \beta) = \Phi(x_i'\beta)$ continuous in β for every realization of x_i. Condition (v) is assumption (b). Since all five conditions are satisfied, Problem 1 implies the NLLS estimator is consistent for β_0. □

Problem 3. *For one dimensional CDFs H and G, the so-called Kolmogorov-Smirnov (KS) metric between the two CDFs is given by*

$$d_{KS}(H, G) := \|H - G\|_\infty = \sup_{-\infty < x < \infty} |H(x) - G(x)|.$$

Define random variables

$$W_n = (1 - d)X_n + dY_n, \quad d \sim \textit{Bernoulli}(\tfrac{1}{2}), \quad X_n \sim \textit{Uni}(0, \tfrac{1}{n}), \quad Y_n \sim \textit{Uni}(1, 1 + \tfrac{1}{n})$$

and

$$W = d, \quad d \sim \textit{Bernoulli}(\tfrac{1}{2}).$$

Let F_n and F denote the CDFs of W_n and W, respectively. Does F_n converge to F in distribution? Does F_n converge to F in the KS metric (i.e., $d_{KS}(F_n, F) \to 0$)? As usual, we take the limit $n \to \infty$.

Solution. The limiting random variable W has CDF $F(w) = 0$ for $w < 0$, $F(w) = \frac{1}{2}$ for $0 \leq w < 1$, and $F(w) = 1$ for $w \geq 1$. For $W_n = (1 - d)X_n + dY_n$ where d, X_n, and Y_n are independent, the CDF is $F_n(w) = \frac{1}{2}P(X_n \leq w) + \frac{1}{2}P(Y_n \leq w)$. Computing each piece yields

$$F_n(w) = \begin{cases} 0 & \text{if } w < 0, \\ \frac{nw}{2} & \text{if } 0 \leq w < \frac{1}{n}, \\ \frac{1}{2} & \text{if } \frac{1}{n} \leq w < 1, \\ \frac{1}{2} + \frac{n(w-1)}{2} & \text{if } 1 \leq w < 1 + \frac{1}{n}, \\ 1 & \text{if } w \geq 1 + \frac{1}{n}. \end{cases}$$

For convergence in distribution, check pointwise convergence at continuity points of F (all $w \notin \{0, 1\}$). For $w < 0$, $F_n(w) = 0 = F(w)$ for all n. For $0 < w < 1$, when n is large enough that $\frac{1}{n} < w$, $F_n(w) = \frac{1}{2} = F(w)$. For $w > 1$, when n is large enough that $w > 1 + \frac{1}{n}$, $F_n(w) = 1 = F(w)$. Therefore $F_n(w) \to F(w)$ at all continuity points of F, so $F_n \xrightarrow{d} F$.

For the Kolmogorov-Smirnov metric, compute $d_{\text{KS}}(F_n, F) = \sup_w |F_n(w) - F(w)|$. For $w \in (0, \frac{1}{n})$, $|F_n(w) - F(w)| = |\frac{nw}{2} - \frac{1}{2}| = \frac{1}{2} - \frac{nw}{2}$, which approaches $\frac{1}{2}$ as $w \to 0^+$. For $w \in (1, 1 + \frac{1}{n})$, $|F_n(w) - F(w)| = |\frac{1}{2} + \frac{n(w-1)}{2} - 1| = \frac{1}{2} - \frac{n(w-1)}{2}$, which approaches $\frac{1}{2}$ as $w \to 1^+$. Therefore $d_{\text{KS}}(F_n, F) = \frac{1}{2}$ for all n. The jumps at 0 and 1 in F create persistent discrepancies of magnitude $\frac{1}{2}$ despite the mass concentrating at these points. Thus F_n converges to F in distribution but not in the KS metric. □

Problem 4. *Suppose we have a consistent estimator $\hat{\mu}$ for $\mu_0 \in \mathbb{R}$, which is also asymptotically normal:*

$$\sqrt{n}(\hat{\mu} - \mu_0) \xrightarrow{d} N(0, \sigma^2).$$

Define

$$\theta_0 = [\max(0, \mu_0)]^2$$

and its estimator

$$\hat{\theta} = [\max(0, \hat{\mu})]^2.$$

Obtain the asymptotic distribution of $\sqrt{n}(\hat{\theta} - \theta_0)$ (you need to classify cases depending on the value of μ_0).

Solution. Define $g(\mu) = [\max(0, \mu)]^2$ and consider three cases based on μ_0. Case 1: $\mu_0 > 0$. In a neighborhood of μ_0, the function g reduces to μ^2 with derivative $g'(\mu_0) = 2\mu_0$. By the delta method, $\sqrt{n}(\hat{\theta} - \theta_0) = \sqrt{n}(g(\hat{\mu}) - g(\mu_0)) \xrightarrow{d} N(0, [g'(\mu_0)]^2\sigma^2) = N(0, 4\mu_0^2\sigma^2)$.

Case 2: $\mu_0 < 0$. Here $g(\mu) = 0$ in a neighborhood of μ_0 since $\max(0, \mu) = 0$ for $\mu < 0$. Thus $\theta_0 = 0$ and $g'(\mu_0) = 0$. For large n, $\hat{\mu} < 0$ with high probability by consistency, implying $\hat{\theta} = [\max(0, \hat{\mu})]^2 = 0$ with probability approaching one. Therefore $\sqrt{n}(\hat{\theta} - \theta_0) = 0 \xrightarrow{p} 0$, giving a degenerate limiting distribution at zero.

Case 3: $\mu_0 = 0$. At the boundary, examine the derivatives of $g(\mu) = [\max(0, \mu)]^2$. The right derivative is $g'_+(0) = \lim_{h \to 0^+} \frac{h^2}{h} = 0$ and the left derivative is $g'_-(0) = \lim_{h \to 0^-} \frac{0}{h} = 0$. Since $g'_+(0) = g'_-(0) = 0$, the function g is differentiable at $\mu_0 = 0$ with $g'(0) = 0$. By the delta method, $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, [g'(0)]^2\sigma^2) = N(0, 0)$, so $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{p} 0$.

Since the first-order term vanishes, the appropriate scaling is n. Note that $\theta_0 = 0$ and $\hat{\theta} = \hat{\mu}^2 \mathbf{1}\{\hat{\mu} \geq 0\}$. Since $n\hat{\theta} = [\sqrt{n} \max(0, \hat{\mu})]^2 = [\max(0, \sqrt{n}\hat{\mu})]^2$, and $\sqrt{n}\hat{\mu} \xrightarrow{d} Z \sim N(0, \sigma^2)$, the continuous mapping theorem yields $n\hat{\theta} \xrightarrow{d} [\max(0, Z)]^2$. The limiting distribution $[\max(0, Z)]^2$ has point mass $P(Z \leq 0) = \frac{1}{2}$ at zero and is otherwise σ^2 times a chi-squared distribution with one degree of freedom, censored at zero.

In sum, $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, 4\mu_0^2\sigma^2)$ when $\mu_0 > 0$, $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{p} 0$ when $\mu_0 \leq 0$, and $n(\hat{\theta} - \theta_0) \xrightarrow{d} [\max(0, Z)]^2$ where $Z \sim N(0, \sigma^2)$ when $\mu_0 = 0$. □

Problem 5. *Suppose we are interested in estimating a parametric nonlinear IV regression model*

$$y = m(x, \theta_0) + \epsilon, \quad \theta_0 \in \Theta \subset \mathbb{R}^k$$

where $E[\epsilon|z] = 0$. Here z is an $\mathbb{R}^k$-valued random vector of instrumental variables. Ideally, we observe a random sample of size n:

$$\{y_i, x_i, z_i\}_{i=1}^n,$$

then we can estimate θ by a non-linear IV (NLIV) estimator that solves

$$\frac{1}{n} \sum_{i=1}^n z_i(y_i - m(x_i, \hat{\theta})) = 0.$$

(i) *Suppose the NLIV estimator is already shown to be consistent. Obtain its asymptotic distribution as $n \to \infty$ (state additional assumptions you wish to make).*

(ii) *Now suppose the econometrician observes* $\{y_i, z_i\}_{i=1}^{n_1}$ *and* $\{x_i, z_i\}_{i=n_1+1}^{n_1+n_2}$ *where* $n = n_1 + n_2$*. A two-sample NLIV (TSNLIV) estimator solves*

$$\frac{1}{n_1}\sum_{i=1}^{n_1} z_i y_i = \frac{1}{n_2}\sum_{i=n_1+1}^{n_1+n_2} z_i m(x_i, \tilde{\theta}).$$

Suppose $n_1 \to \infty$*,* $n_2 \to \infty$*, and* $\frac{n_1}{n_2} \to k > 0$*. Suppose* $\tilde{\theta}$ *is consistent. Obtain its asymptotic distribution.*

Solution. Part (i): Assume standard regularity conditions including interiority of θ_0, smoothness of $m(x,\theta)$ with continuous derivative $\nabla_\theta m(x,\theta)$, moment condition $E[z_i\epsilon_i] = 0$, finite moment $E[\|z_i \nabla_\theta m(x_i,\theta_0)\|] < \infty$, and full rank of $G := E[z_i \nabla_\theta m(x_i,\theta_0)']$. Define $\Omega := E[z_i z_i' \epsilon_i^2]$, assumed to exist and be positive definite. The NLIV estimator $\hat{\theta}$ satisfies $\frac{1}{n}\sum_{i=1}^n z_i(y_i - m(x_i,\hat{\theta})) = 0$. Since $\hat{\theta} \xrightarrow{p} \theta_0$ and $y_i = m(x_i,\theta_0) + \epsilon_i$, expand around θ_0 to obtain $0 = \frac{1}{n}\sum_{i=1}^n z_i\epsilon_i - \frac{1}{n}\sum_{i=1}^n z_i \nabla_\theta m(x_i,\theta_0)'(\hat{\theta} - \theta_0) + o_p(\|\hat{\theta} - \theta_0\|)$. Multiplying by $\sqrt{n}$ yields $\left[\frac{1}{n}\sum_{i=1}^n z_i \nabla_\theta m(x_i,\theta_0)'\right]\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^n z_i\epsilon_i + o_p(1)$. By the law of large numbers, $\frac{1}{n}\sum_{i=1}^n z_i \nabla_\theta m(x_i,\theta_0)' \xrightarrow{p} G$. By the central limit theorem, $\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i\epsilon_i \xrightarrow{d} N(0,\Omega)$. By Slutsky's theorem, $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, G^{-1}\Omega G'^{-1})$.

Part (ii): The TSNLIV estimator $\tilde{\theta}$ solves $\frac{1}{n_1}\sum_{i=1}^{n_1} z_i y_i = \frac{1}{n_2}\sum_{i=n_1+1}^{n_1+n_2} z_i m(x_i,\tilde{\theta})$. Since $E[z_i y_i] = E[z_i m(x_i,\theta_0)] =: \mu$, subtract μ from both sides to get

$$\frac{1}{n_1}\sum_{i=1}^{n_1} z_i\epsilon_i = \frac{1}{n_2}\sum_{i=n_1+1}^{n_1+n_2} z_i[m(x_i,\tilde{\theta}) - m(x_i,\theta_0)].$$

Expanding around θ_0 yields $\frac{1}{n_1}\sum_{i=1}^{n_1} z_i\epsilon_i \approx \frac{1}{n_2}\sum_{i=n_1+1}^{n_1+n_2} z_i \nabla_\theta m(x_i,\theta_0)'(\tilde{\theta} - \theta_0)$. Multiplying by $\sqrt{n_2}$ gives

$$\left[\frac{1}{n_2}\sum_{i=n_1+1}^{n_1+n_2} z_i \nabla_\theta m(x_i,\theta_0)'\right]\sqrt{n_2}(\tilde{\theta} - \theta_0) = \sqrt{n_2}\cdot\frac{1}{n_1}\sum_{i=1}^{n_1} z_i\epsilon_i + o_p(1).$$

Define the vector

$$V_n = \begin{pmatrix} \frac{1}{\sqrt{n_1}}\sum_{i=1}^{n_1} z_i\epsilon_i \\ \frac{1}{n_2}\sum_{i=n_1+1}^{n_1+n_2} z_i \nabla_\theta m(x_i,\theta_0)' \end{pmatrix}.$$

Since the two samples are independent, by CLT and LLN, $V_n \xrightarrow{d} \begin{pmatrix} N(0,\Omega) \\ G \end{pmatrix}$, where $G = E[z_i \nabla_\theta m(x_i,\theta_0)']$ and $\Omega = E[z_i z_i' \epsilon_i^2]$.

Since $\sqrt{n_2}(\tilde{\theta} - \theta_0) = h(V_n) + o_p(1)$ where $h\begin{pmatrix} u \\ M \end{pmatrix} = M^{-1}\cdot\sqrt{n_2/n_1}\cdot u$ and $n_2/n_1 \to 1/k$, the continuous mapping theorem yields $\sqrt{n_2}(\tilde{\theta} - \theta_0) \xrightarrow{d} G^{-1}\cdot\frac{1}{\sqrt{k}}\cdot N(0,\Omega) = N\left(0, \frac{1}{k}G^{-1}\Omega G'^{-1}\right)$.

Since $n/n_2 \to (1+k)$, we have $\sqrt{n}(\tilde{\theta} - \theta_0) = \sqrt{n/n_2}\cdot\sqrt{n_2}(\tilde{\theta} - \theta_0) \xrightarrow{d} N\left(0, \frac{1+k}{k}G^{-1}\Omega G'^{-1}\right)$. The asymptotic variance is inflated by factor $(1 + 1/k)$ relative to the full-sample NLIV estimator. When $n_1 = n_2$ (so $k = 1$), the variance doubles; when $n_2 \ll n_1$ (small k), the variance inflation is substantial because the second sample, which contains the covariate information, is relatively small. □

Problem 6. *Recall the quantile IV model discussed in class, though here we consider a case with a parametric assumption. Let τ fixed in $(0,1)$ throughout this question. Let (y,x,z) be an observable random vector taking its value in $\mathbb{R}\times\mathbb{R}^k\times\mathbb{R}^q$. Suppose a linear model for y in terms of x, which is endogenous, satisfies the following quantile IV condition at the τ-th quantile*

$$E[\mathbf{1}\{y \le x'\theta_0\} - \tau | z] = 0, \quad \theta_0 \in \Theta \subset \mathbb{R}^k$$

where z is interpreted as instruments. Let Θ be compact. We assume that θ_0 is a unique solution to

$$E\left[z\left(\mathbf{1}\{y \le x'\theta_0\} - \tau\right)\right] = 0.$$

Suppose we observe a random sample $\{(y_i,x_i,z_i)\}_{i=1}^n$. Define

$$Q_n(\theta) = \bar{g}_n(\theta)' W_n \bar{g}_n(\theta)$$

where

$$\bar{g}_n(\theta) = \frac{1}{n}\sum_{i=1}^n z_i\left(\mathbf{1}\{y_i \le x_i'\theta\} - \tau\right)$$

and W_n is a $q\times q$ weighting matrix with $W_n \xrightarrow{p} W$ where W is positive definite. Here we assume that $\Pr\{(y,x')'c = 0\} = 0$ for every non-zero vector $c \in \mathbb{R}^{k+1}$. Finally, assume that $E[|z|] < \infty$, where $|z|$ denotes the vector of the absolute values of the elements of z.

Show that the quantile IV estimator that minimizes Q_n over Θ is consistent by verifying appropriate (low level) sufficient conditions.

Solution. The quantile IV estimator minimizes $Q_n(\theta) = \bar{g}_n(\theta)'W_n\bar{g}_n(\theta)$ where $\bar{g}_n(\theta) = n^{-1}\sum_{i=1}^n z_i(\mathbf{1}\{y_i \le x_i'\theta\} - \tau)$. Define the population moment function $g(\theta) = E[z(\mathbf{1}\{y \le x'\theta\} - \tau)]$ and population objective $Q(\theta) = g(\theta)'Wg(\theta)$. First, verify that θ_0 uniquely minimizes $Q(\theta)$. By the conditional moment restriction $E[\mathbf{1}\{y \le x'\theta_0\} - \tau|z] = 0$, taking expectations yields $g(\theta_0) = E[z(\mathbf{1}\{y \le x'\theta_0\} - \tau)] = 0$, hence $Q(\theta_0) = 0$. For any $\theta \ne \theta_0$, the uniqueness assumption implies $g(\theta) \ne 0$. Since W is positive definite, $Q(\theta) = g(\theta)'Wg(\theta) > 0$ for all $\theta \ne \theta_0$. Therefore θ_0 uniquely minimizes $Q(\theta)$ over Θ.

Next, establish uniform convergence. The parameter space Θ is compact by assumption. The indicator function $\mathbf{1}\{y \le x'\theta\}$ is discontinuous at points where $y = x'\theta$, but the assumption $\Pr\{(y,x')'c = 0\} = 0$ for all nonzero c ensures that for each θ, the set of discontinuity points has probability zero. Since $\mathbf{1}\{y \le x'\theta\} - \tau$ takes values in $[-\tau, 1-\tau]$ and $E[|z|] < \infty$, have $E[\|z(\mathbf{1}\{y \le x'\theta\} - \tau)\|] \le E[\|z\|] < \infty$ uniformly in θ.

The class of functions $\mathcal{F} = \{\mathbf{1}\{y \le x'\theta\} : \theta \in \Theta\}$ is a Vapnik-Chervonenkis class with VC dimension at most $k+1$ (corresponding to half-spaces in $\mathbb{R}^{k+1}$). Combined with the envelope condition $E[|z|] < \infty$ and compactness of Θ, the uniform law of large numbers for VC-type classes (e.g., by Talagrand's inequality) yields $\sup_{\theta\in\Theta}\|\bar{g}_n(\theta) - g(\theta)\| \xrightarrow{p} 0$. Since $W_n \xrightarrow{p} W$ and the quadratic form $(\xi, M) \mapsto \xi'M\xi$ is continuous, $\sup_{\theta\in\Theta}|Q_n(\theta) - Q(\theta)| = \sup_{\theta\in\Theta}|\bar{g}_n(\theta)'W_n\bar{g}_n(\theta) - g(\theta)'Wg(\theta)| \xrightarrow{p} 0$.

By the consistency theorem for extremum estimators, since $Q_n(\theta)$ converges uniformly to $Q(\theta)$ over the compact parameter space Θ and θ_0 uniquely minimizes $Q(\theta)$, the quantile IV estimator $\hat{\theta}_n = \arg\min_{\theta\in\Theta} Q_n(\theta)$ satisfies $\hat{\theta}_n \xrightarrow{p} \theta_0$. Therefore the quantile IV estimator is consistent for θ_0. □